



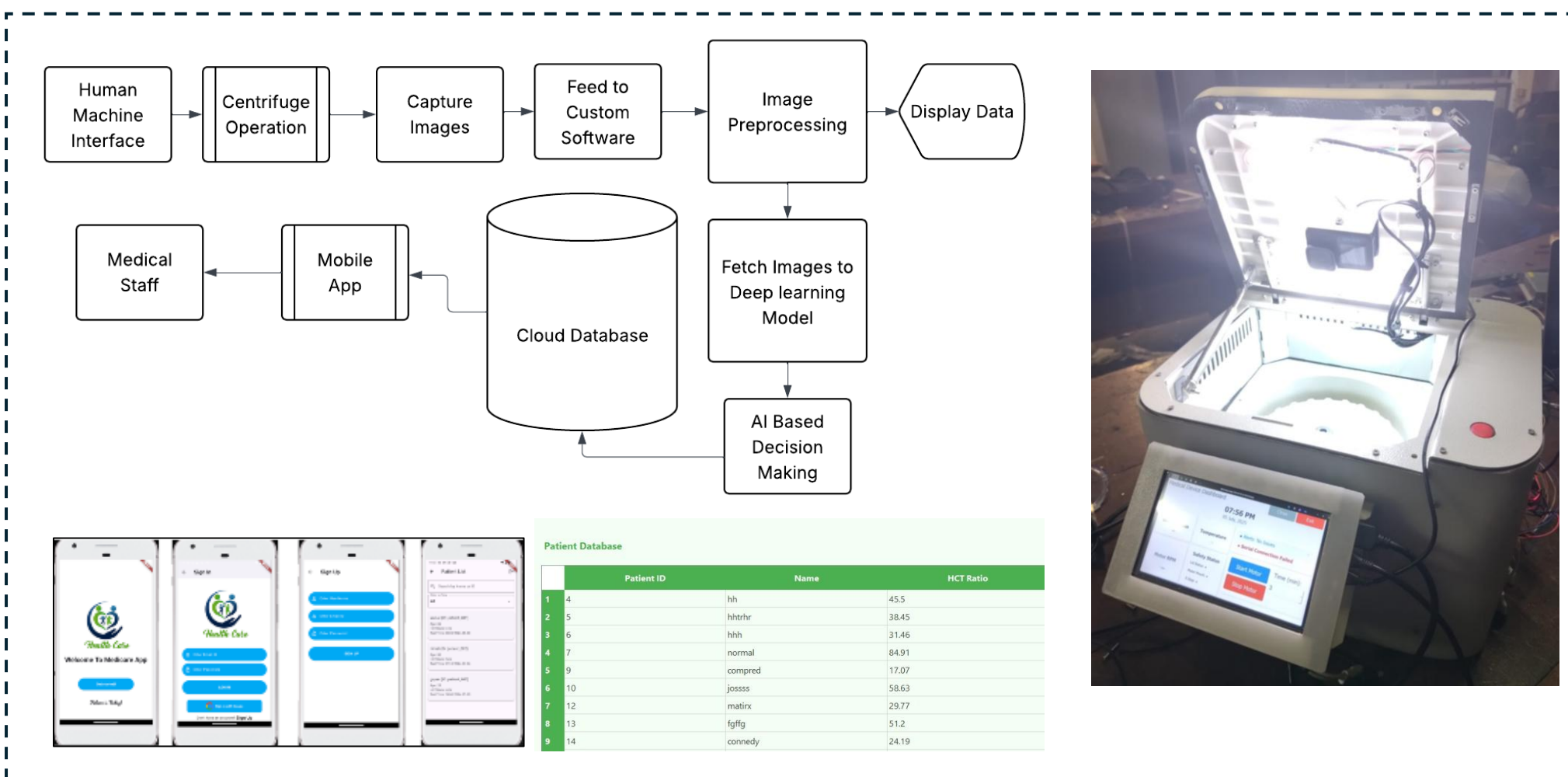
Novel Vision Based PCV Analyzer for Efficient Hematocrit Monitoring in Dengue Patients

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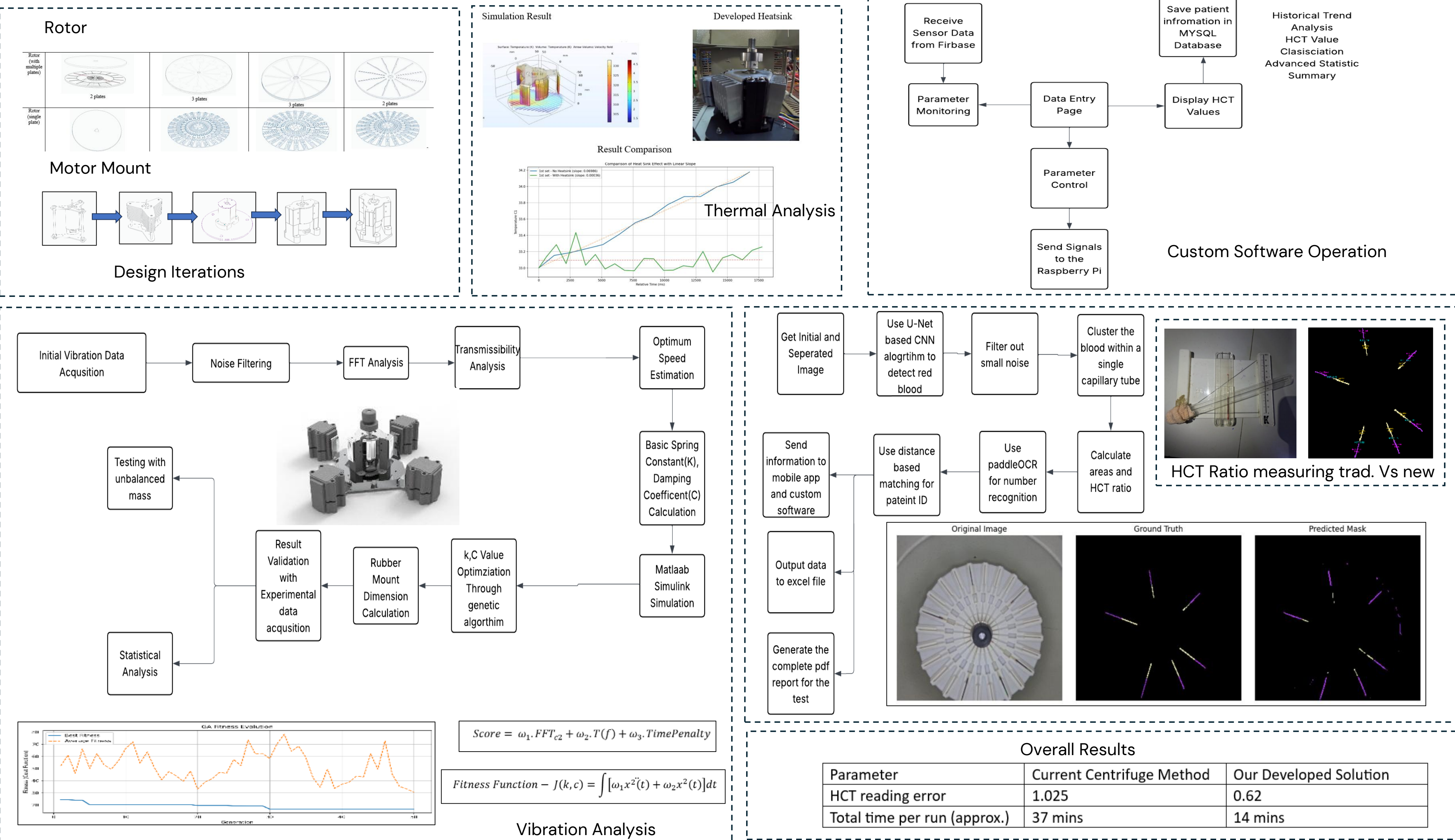
Introduction & Background

Blood consists of red blood cells and plasma. The Hematocrit (HCT) ratio is simply the ratio between the red blood to the total blood volume. The packed cell volume test is conducted using a centrifuge machine to separate the blood, and then the HCT ratio is measured, which is a critical parameter for dengue patient monitoring. After the centrifugation, the current process uses a 'reader card' approach to get the HCT value, which is very time-consuming and prone to human error. This project develops a novel solution by using **vision-based image processing** techniques to eliminate human error and greatly increase efficiency. The developed centrifuge consists of a **single plate rotor with a self-locking mechanism** to keep the tubes intact during rotation. The centrifugation speed and time were selected through rigorous experimental analysis. Further, instead of manually recording the data, **IoT** is used to send the data to a **mobile app** to all relevant medical staff. The custom-developed software can be used for laboratory analysis to conduct future analysis, such as classifying HCT ratios and using it in HCT value prediction.

System Overview



Methodology & Results



Conclusion & Future Work

- ❖ The current algorithm for Image processing has improved the accuracy and works fine in the current environment and the lightning. But by varying those parameters and retraining a more universal dataset, the algorithm can be implemented even in other devices to automate the HCT value calculation. The use of a vision transformer can increase the accuracy even higher for this scenario with high variability.
- ❖ The HCT analysis and prediction done in the custom software are not sufficiently accurate to use in the medical environment. The accuracy can be increased by using a real dataset of patients and the diagnostic results. These values can be used for a fuzzy C-means clustering algorithm to refine the membership functions and give more realistic inputs to use the HCT value prediction.
- ❖ Integrate the mobile app with the patient database in the hospital and connect all relevant users.